

BORROWED SIZE OF SMALL AND MEDIUM CITIES IN A HIERARCHICAL URBAN SYSTEM

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ABSTRACT

Small and medium-sized cities can benefit from their proximity to large cities through borrowed size and spillover effects. First, geographical proximity may facilitate these effects through spatial diffusion. The closer small- and medium-sized cities are geographically to larger cities, the more likely they are to experience borrowed size effects. Second, hierarchical proximity—being closer in status within the urban system—can also lead to borrowed size benefits, regardless of their size. These effects can be achieved through contagious and hierarchical diffusion, respectively. Focusing on Korea's strongly hierarchical urban system, the study analyses 152 small- and medium-sized cities using 8 indices and regression models. The results reveal that both geographic closeness to the primate city and a higher hierarchical position in the national urban system are key drivers of borrowed size. This study contributes to the discussion on agglomeration economies and network effects in urban economic development.

Key words: Small- and medium-sized cities; borrowed size; geographical proximity; hierarchical proximity; strong urban hierarchy; Korean urban system

INTRODUCTION

Cities interact with each other in space. While some city pairs are more equal in their interactions, others reveal more hierarchical relationships. In the latter case, the relationship is mostly one-directional and geographic proximity to large cities is usually a penalty rather than an advantage in terms of city growth. However, an interesting observation has been reported in many studies, mainly on European cities, showing that small cities can benefit from large adjacent cities. One explanation is the weakened hierarchical structure of the European urban system, which consists of historically independent and relatively similar cities in size (Cardoso & Meijers 2016). If two cities are more similar in size, their relations and, consequently, their interactions are more likely to be equal (i.e. providing mutual benefit to each other).

The positive effects that small cities can receive from large cities have been formalised using the concept of borrowed size effects. According to Alonso (1973, p. 200), the concept of borrowed size represents a phenomenon that a 'small city or metropolitan area exhibits some of the characteristics of a larger one if it is near other population concentrations'. This can be considered the spillover effect from large cities, through which small cities perform better than their predicted population sizes (Burger *et al.* 2015). Small cities near large cities can benefit from agglomeration effects due to proximity, while avoiding agglomeration costs (Phelps *et al.* 2001; Veneri & Burgalassi 2012).

The proximity of small cities to large cities in generating borrowed size effects can be viewed in two ways. First, geography or geographical proximity has these effects. If

it works in this way, the presence of the effects in small- and medium-sized cities can be more prominent if they are geographically closer to large cities. In this case, geographical proximity can be a source of borrowed size effects. Second, a city's hierarchical position in the urban system or hierarchical proximity can also generate these effects. If geographical proximity reflects a short distance between small and large cities in a geographical space, hierarchical proximity refers to a short distance between two types of cities in an urban hierarchical space. In other words, cities with short distances from large cities in the urban hierarchy are those with smaller gaps from large cities in terms of rank or size, that is, larger small cities. If this is the case, borrowed size effects may be more prominent in cities closer to large ones in the urban hierarchy (rather than simply proportionate to a city's ranking or size in the urban system). In this case, hierarchical proximity could be the source of borrowed size effects. These two types of proximity effects can also be associated with traditional diffusion typologies. The borrowed size effects in the former setting can be achieved through contagious diffusion, while those in the latter may be achieved through hierarchical diffusion.

What happens when an urban system has a strong hierarchy? Do large cities have a stronger influence on smaller cities and are small cities expected to be more subordinate to large cities? Do small cities experience stronger penalties than benefits, leading to a greater gap between cities of different sizes? More precisely, in a national urban system characterised by strong hierarchical structures, it remains unresolved whether the principal source of borrowed size effects lies in large cities within the same functional urban region or in the primate city (the capital). Likewise, from the perspective of hierarchical proximity, no clear consensus has been reached on whether hierarchical proximity at the national level or at the regional level is more significant. Considering that most studies in this area focus on European cities, where the polycentric urban system is more common, examining East Asian urban systems, which are characterised by a stronger hierarchical structure and a greater role

of government intervention, can provide new insights into the relationship between large and small cities. This will also provide an opportunity for a more generalised understanding of the relationship between small- and medium-sized cities and large cities, and consequently, the urban dynamics between cities of different sizes (e.g. Meijers & Sandberg 2008; Cardoso & Meijers 2016).

To pursue a more comprehensive and accurate understanding of the relationships between small and large cities in an urban system with a strong hierarchy, this study explores the presence and dynamics of borrowed size effects in the Korean urban system. More specifically, this study attempts to examine whether these two proximity factors are influential in driving borrowed size effects, and if so, which of these two factors exerts a stronger influence on the urban system characterised by the dominance of a few large cities in Korea. To reveal the dynamics of these effects, this study emphasises two aspects: borrowed functions and performance (Burger *et al.* 2014; Burger *et al.* 2015). For this study, eight indices related to functions and performance were derived from earlier studies and used as response variables in the regression model for 152 small- and medium-sized cities in Korea. By examining the economic dynamics of small- and medium-sized cities in the context of Korea, this study is expected to contribute to recent debates on the effects of agglomeration and network economy on urban economic development.

AGGLOMERATION SHADOWS AND BORROWED SIZES ON THE NETWORK OF CITIES

The agglomeration shadow is a concept rooted in New Economic Geography, which explains why cities near larger ones cannot grow due to the competition effect (Dobkins & Ioannides 2001; Meijers *et al.* 2016). Arthur (1990, p. 247) first noted this concept when explaining the Löschian spatial system of spreading and separation: 'Locations with large numbers of firms therefore cast an 'agglomeration shadow' in which little or no settlement takes place. This causes separation of the industry'.

This implies that agglomeration economies can cause spreading and separation, in addition to concentration. If such agglomeration power is strong, urban systems bifurcate into the core and periphery; the latter is a result of the agglomeration shadow effect (Ikeda *et al.* 2017; Li & Sun 2020). In such cases, the growth and development of the core city occurs through the sacrifice of the peripheral city in the hinterland. From the perspective of a peripheral city, the agglomeration shadow effect occurs if it has fewer functions than expected based on the city's size. This effect is typically associated with the negative effects of network connections between cities (Meijers *et al.* 2016). As Volkmann and Rusche (2020) note, empirical evidence of agglomeration shadows is observed in many places in Western Europe, including Northwest Europe (Burger *et al.* 2015), the Netherlands and Germany (Meijers 2008). East Asian countries, which have a stronger hierarchy in the urban system, also experience this effect (Yang *et al.* 2022).

The borrowed size effect is the opposite of the agglomeration shadow effect. This explains why cities near larger cities can grow due to the synergy effect. While the agglomeration shadow reflects spatial competition effects, borrowed size is related to the effect of the network economy between cities (Burger *et al.* 2015). The concept of borrowed size was first introduced by Alonso (1973, p. 200): 'the concept of a system of cities has many facets, but one of particular interest for our topic is the concept of borrowed size, whereby a small city or metropolitan area exhibits some of the characteristics of a larger one if it is near other population concentrations'. Small- and medium-sized cities may be more advantageous if there is a borrowed size effect. They can avoid the cost of agglomeration diseconomies by maintaining their small size while enjoying the benefits of agglomeration economies by staying near larger cities, which leads to a balance between agglomeration benefits and costs (Phelps *et al.* 2001; Meijers & Burger 2017).

Borrowed size suggested by Alonso (1973) seems to imply agglomeration externality fields as a variation of classical Marshallian externalities (Phelps *et al.* 2001; Burger & Meijers 2016). With advances in transportation and communication, not only have innovation

and knowledge spillovers expanded, but human mobility has also extended over longer distances, leading agglomeration economies to evolve from being nodal to increasingly zonal in nature (Phelps 2004). According to Solis *et al.* (2022), the effect of borrowed size can be realised through Marshallian-type sharing, matching and learning from networks of nearby cities. By combining the concepts of borrowed size and agglomeration economies in a more explicit manner, especially within the metropolitan area of a one-hour travel distance, Polese and Shearmur (2006) call the effect borrowed agglomeration economies.

Some scholars believe that borrowed size is not a concept that actually occurs. Hans and Koster (2018) suggest that borrowed size is simply an empirical issue resulting from the mismatch between administrative spatial units and functional areas. Anselin *et al.* (2000) note that urbanisation economies can reach beyond the geographical boundary of the administrative spatial units. However, considering that a city's economic policies are usually created at the level of administrative spatial units, the relationships between nearby cities still need to be treated with importance. Furthermore, it remains unclear why some pairs of nearby administrative spatial units have relationships based on borrowed size, whereas others have relationships based on agglomeration shadows.

Subsequently, Meijers and Burger (2017) extended the concept of borrowed size by arguing that it is not merely a matter of smaller cities benefiting from larger ones or arising solely from geographical proximity. Rather, interactions within networks of cities across multiple spatial scales may also constitute an important source of borrowed size. In this view, whether a city exhibits an agglomeration shadow or a borrowed size effect in its relationship with nearby larger cities depends on the nature of the relationship—competition or collaboration—which again depends on the existence of functional complementarity. If it is the latter, the network serves as a substitute for agglomeration benefits through interaction within the system of cities (Meijers & Burger 2017).

The borrowed size concept can further be decomposed into borrowed function and borrowed performance; if both effects occur, it is called borrowed size (Meijers & Burger 2017;

Volgmann & Rusche 2020; Zhao *et al.* 2025). With this extension, the borrowed size can be redefined more accurately. According to Meijers and Burger (2017), borrowed size occurs when a city exhibits urban functions and/or performance as an output of interactions within the city network through relations with typically larger cities at various spatial scales, in which agglomeration benefits can be substituted. This highlights a distinctive process of borrowed size in which the growth or specific functions of higher-tier cities exert unilateral positive effects on lower-tier cities. We then argue that hierarchical proximity can play a crucial role in this process of generating benefits in a more effective way from network externalities for lower-tier cities.

This hierarchical proximity can operate in three major ways within borrowed size effects. First is traditional hierarchical diffusion. Cultural facilities, specialised services and knowledge or information located in large cities spread to surrounding cities through hierarchical network connections such as transport, information flows and inter-institutional cooperation, and they are transmitted first to

the next tier of cities (Pedersen 1970). Second, the geographical spacing between hierarchically proximate cities reduces the level of spatial competition, allowing them to borrow the size of surrounding areas more effectively. Third, political effects emerge in the process of national resource allocation. Even in countries with strong capital dominance, resources cannot be concentrated exclusively in the capital. Smaller cities with limited population bases still receive essential public services—such as education and healthcare—through policy interventions. This is referred to as ‘institutional borrowed size’ (Meijers & Burger 2017). Typically, the resources are distributed to upper-tier cities rather than lower-tier ones in order to pursue balanced national development in a more effective way through selection and concentration (Zhang *et al.* 2020; Shi *et al.* 2025). This suggests that the hierarchical proximity to the top-tier city is important in obtaining institutional borrowed size effects. Figure 1 provides a visual illustration of geographic proximity, hierarchical proximity and network connectivity in relation to the borrowed size effect.

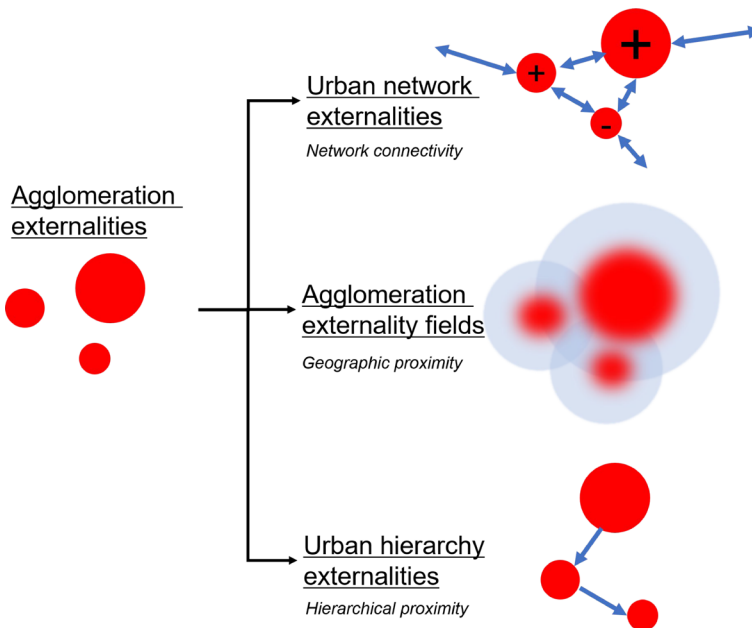


Figure 1. Possible spatial transformations of agglomeration externalities. Source: Adapted from Burger and Meijers (2016, p. 8), with ‘urban hierarchy externalities’ category added by the author.

FACTORS INFLUENCING BORROWED SIZE EFFECTS

Several factors can influence whether a pair of cities has a collaborative relationship, resulting in the occurrence of borrowed size effect. This review focuses on factors related to geographic aspects and urban system characteristics. The first factor is distance (from the core city). Studies have found that improvements in accessibility lead to a stronger agglomeration shadow near a larger city, while the borrowed size effect can only be realised at a certain distance from the city (Wang *et al.* 2022b; Tian *et al.* 2023).

Some studies have suggested that physical distance is no longer that important in determining the type of relationship a city has with other cities (de Vos *et al.* 2020; Otsuka 2020). For example, Hesse (2016) points out that the determining factor is whether the city belongs to a network of polycentric urban regions and the inter-urban networks between polycentric urban regions. A transportation network can facilitate the opportunity to receive the benefits derived from network membership (Otsuka 2024, 2025; Zhao *et al.* 2025).

City size is also a determining factor in this relationship. Some studies suggest that both larger and smaller cities experience borrowed size effects, but the types of effects vary depending on city size. Larger cities generally borrow functions from smaller cities by expanding their markets to surrounding smaller cities, while smaller cities borrow performances from larger cities (Meijers & Burger 2017; Wang *et al.* 2022a).

The size of the partner city also influences the type of relationship. The basic idea of borrowed size, introduced by Alonso (1973), assumes that the benefit obtained by smaller cities is related to their relationships with larger cities. However, more empirical findings suggest that the borrowed size effect occurs in the opposite direction. For example, Sohn *et al.* (2022) argue that borrowed size effects occur between cities of similar size, whereas larger cities only produce agglomeration shadow effects on adjacent smaller cities. This means that larger cities perhaps are more likely to borrow size from smaller cities than vice versa. Burger *et al.* (2015) suggest that a city's

position in the city network or system of cities determines whether the relationship between cities is based on borrowed size or the agglomeration shadow effect.

To some extent, city size is not the most important factor; other significant factors must be considered. For example, Otsuka (2020) notes that borrowed size effects can only be realised in horizontal and non-hierarchical relationships. Burger *et al.* (2015) point out that whether it relates to borrowed size or the agglomeration shadow effect depends on the complementarity of the functions between cities. Combining these two arguments suggests that borrowed size effects are more likely to occur if a pair of cities has horizontal and/or non-hierarchical relationships and if they are complementary to each other in urban functions.

Metropolitan geography or urban spatial structure (monocentricity vs. polycentricity) is important for determining the effect of borrowed size. Specifically, Volkmann and Rusche (2020) note that urban spatial structure, along with city size, has a significant influence on the positioning of a city among the four dimensions of borrowed size types (borrowed size, borrowed performance, borrowed function and agglomeration shadow). Several studies have suggested that this effect is associated with polycentric urban regions. Polycentric metropolitan areas provide functional interconnections and proximity (Meijers & Burger 2017). Phelps *et al.* (2001) link this effect to financial and technological externalities. However, such a positive functional spillover effect across a city boundary can be facilitated only when the city cluster has a large population size and good connectivity of urban infrastructure (Wang *et al.* 2022a). Meijers and Van Der Wouw (2019) extend the spatial boundary of the borrowed size effect from a metropolitan area to a larger spatial scale city network under the condition that accessibility or network connections to metropolitan agglomeration externalities are satisfied. Similarly, Burger *et al.* (2015) note that good accessibility enables the borrowed size effect to extend to national or international spatial scales, although the typical spatial scale of this effect is limited to functional urban regions with strong interactions among

adjacent cities. It is interesting to note a study showing that monocentricity is more effective in activating the borrowed size effect, contrary to general intuition. Sun *et al.* (2019) report that morphological monocentricity generates borrowed size effects through labour mobility, whereas polycentric regions do not borrow functions or performance, as anticipated. These findings suggest that while the borrowed size effect is more often observed in polycentric urban regions, it is important to understand why it still occurs in some monocentric urban regions.

Although most studies on the forms and mechanisms of borrowed size have focused on European cities, recent research on Asian countries, particularly China and Japan, has expanded the empirical scope beyond the Eurocentric perspective and offered new insights into the phenomenon. Similar to the European findings, studies in Asian contexts have provided empirical evidence that the borrowed size effect tends to outweigh the agglomeration shadow effect (Gong & Zhong 2021; Wang *et al.* 2022a; Yang *et al.* 2024; Zhao *et al.* 2025), and have emphasised the importance of interregional transportation infrastructure in enhancing productivity through borrowed size (Otsuka 2024, 2025). However, most of these studies have primarily focused on geographical proximity and connectivity among cities within the same region, paying less attention to the role of national and regional urban hierarchies in shaping borrowed size effects. In national urban systems where strong hierarchical structures prevail, it is still unclear whether borrowed size effects primarily originate from large cities within the same functional urban region or from the primate city, typically the capital. Likewise, it remains unresolved which factor among geographical proximity, hierarchical proximity and network connectivity is more effective in generating borrowed size. In the case of hierarchical proximity, there is also no consensus on whether its influence is more pronounced at the national scale or at the regional scale. This study seeks to examine, in the context of Korea's strongly hierarchical urban system, the effects of geographical and hierarchical proximity at both the national and regional scales on borrowed size.

RESEARCH METHODOLOGY

Defining small- and medium-sized cities in Korea –

Figure 2 shows the Provinces in Korea, along with all the cities included in the analysis. Seoul, located in the northwestern part of Korea, is the country's dominant economic centre, as well as its political capital. Six Metropolitan Cities are shown on the map. They have a population of over one million and serve as the regional economic centres of the corresponding provinces. In this study, these seven cities are classified as large cities, whereas the remaining 152 cities are considered small- and medium-sized cities. The largest metropolitan city in a province is considered the regional centre. The location of this city is used later when measuring the distance from small- and medium-sized cities to the regional centre. However, Incheon and Ulsan are not considered regional centres because they are too close to the larger Metropolitan Cities, Seoul and Busan, respectively. Furthermore, since there are no metropolitan cities in Gangwon, Chungcheongbuk and Jeollabuk provinces, the largest cities in the corresponding regions, namely, Wonju, Cheongju and Jeonju, are classified as regional centres. Finally, the analysis only considered contiguous Korea; therefore, two cities on Jeju Island were excluded. Sejong was excluded because of its status as a special self-governing city with a concentration of central government functions.

Data and variables –

In our analysis, we concentrated on two different aspects of the urban characteristics of a larger city: the centre of consumption and production (Glaeser *et al.* 2001). These are closely related to the two dimensions of urban functions and performance. First, we focused on the presence of metropolitan functions. The metropolitan functions were diverse. Previous studies focused on various functions related to ((inter)national) politics, public administration, science, education, business, business services, accessibility, cultural and sports amenities and shopping (Phelps 1998; Phelps *et al.* 2001; Burger *et al.* 2014; Meijers *et al.* 2014; Burger

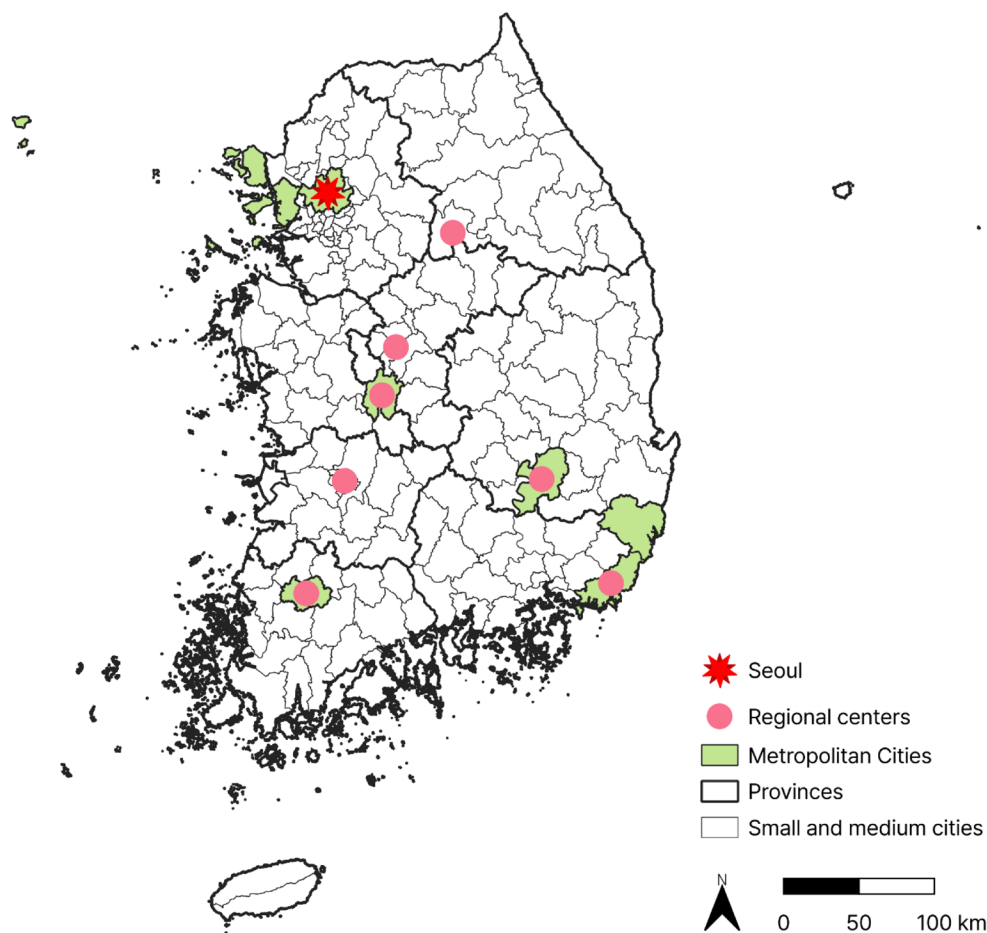


Figure 2. *Small and medium cities and Metropolitan Cities in Korea.*

et al. 2015; Hesse 2016; Meijers *et al.* 2016; Meijers & Burger 2017; Sun *et al.* 2019; de Vos *et al.* 2020; Volgmann & Rusche 2020; Sohn *et al.* 2022). These metropolitan functions can be good for measuring the presence of borrowed size effects as they strongly depend on the size of the city and the surrounding regions (Burger *et al.* 2014). In this study, we selected five categories of metropolitan functions that are frequently used in earlier studies: business, consumption, cultural amenities, transportation and information technology. Variables representing each category were chosen such that they represented functions dominant in metropolitan cities, usually more than their shares of population sizes (see Table 1).

Second, several measures of economic indicators were used as dependent variables for urban performance. Previous studies have identified various aspects of urban performance, including personal income, population growth, employment and its growth, startups, knowledge-based activities, producer services and housing costs (Alonso 1973; de Vos *et al.* 2020; Hans & Koster 2018; Lenaerts *et al.* 2023; Meijers & Burger 2017; Partridge *et al.* 2007; Partridge *et al.* 2008, 2009, 2010; Polese & Shearmur 2006; Volgmann & Rusche 2020; Solis *et al.* 2022). This study summarises these variables into three categories and uses them in the empirical analysis: income, innovation and vitality.

The effects of relationships between large cities and small- and medium-sized cities on urban functions and performance can be explained by several variables. In our analysis, we

considered both hierarchical and geographical measures at national and regional scales.

First, we examined the positions of cities within the urban hierarchy at both scales to

Table 1. *Variable description (N = 152).*

Variables	Descriptions	Sources
Dependent variables		
BUSINESS	The number of employees in advanced producer service sectors	Korean Statistical Information Service (2022)
CONSUME	The number of department stores	Korean Statistical Information Service (2022)
CULTURE	The number of employees in creative and art-related service sectors	Korean Statistical Information Service (2022)
TRANSPORT	The number of passengers using high-speed train stations	Korean Railway Industry Information Center (2022) Public Data Portal (2022)
IT	The number of establishments in information technology sector	Korean Statistical Information Service (2022)
INCOME	Per capita GRDP	Korean Statistical Information Service (2022)
INNOVATE	The number of patents	Intellectual Property Statistics Service (2022)
VITALITY	The number of startups	Organisation for Venture Enterprise Certification (2022)
Independent variables		
Hierarchical position		
SECOND	1 if it is administrative city (but not Metropolitan City)	Author's calculation
THIRD	1 if it is county	Author's calculation
UP_SECOND	1 if it is the upper-side second-tier city in national urban hierarchy	Author's calculation
LOW_SECOND	1 if it is the lower side second-tier city in national urban hierarchy	Author's calculation
UP_THIRD	1 if it is the upper-side third-tier city in national urban hierarchy	Author's calculation
LOW_THIRD	1 if it is the lower side third-tier city in national urban hierarchy	Author's calculation
REG_SECOND	1 if it is the upper-side city in regional urban hierarchy	Author's calculation
REG_THIRD	1 if it is the lower side city in regional urban hierarchy	Author's calculation
REG_UP_SECOND	1 if it is the upper-side second-tier city in regional urban hierarchy	Author's calculation
REG_LOW_SECOND	1 if it is the lower side second-tier city in regional urban hierarchy	Author's calculation
REG_UP_THIRD	1 if it is the upper-side third-tier city in regional urban hierarchy	Author's calculation
REG_LOW_THIRD	1 if it is the lower side third-tier city in regional urban hierarchy	Author's calculation
Geographical position		
DIST_SEOUL	Euclidean distance (km) to Seoul	Author's calculation
DIST_CENTRE	Euclidean distance (km) to regional economic centre	Author's calculation

assess whether a city's hierarchical position affects its level of urban performance and the presence of urban functions, and to determine which scale—national or regional—has greater explanatory power. The dummy variable *SECOND* indicates administrative cities, while *THIRD* represents counties (i.e. neither metropolitan nor administrative cities). To capture the differential effects of urban hierarchy more precisely, we further disaggregated the second- and third-tier cities into four groups in the national urban hierarchy: *UP_SECOND* (upper second-tier cities), *LOW_SECOND* (lower second-tier cities), *UP_THIRD* (upper third-tier cities) and *LOW_THIRD* (lower third-tier cities). Similarly, we identified regional first-tier cities as centres of eight regional urban systems and created six additional dummy variables for each region: *REG_SECOND*, *REG_THIRD*, *REG_UP_SECOND*, *REG_LOW_SECOND*, *REG_UP_THIRD* and *REG_LOW_THIRD*. Because the control group at each scale consists of (lower) third-tier cities, a significant positive coefficient for a specific dummy variable closer to the first-tier city (e.g. *UP_SECOND*) indicates that the borrowed size effects arising from hierarchical proximity to higher-level cities are limited to that specific position(s) (e.g. upper second-tier cities) in the urban hierarchy. Furthermore, if the dummy variable for second-tier cities at the national scale is significant but the corresponding variable at the regional scale is not, it suggests that the national scale is more important; the reverse would indicate greater importance of the regional scale. Classification between upper and lower groups was conducted using the k-means clustering method based on population size.

Second, we evaluated the effects of geographical proximity to first-tier cities at both national and regional scales on urban functions and performance. At the national scale, we employed the distance from Seoul (*DIST_SEOUL*) to examine whether spillover effects from the capital exist. If the external effects diminish as the distance from Seoul increases, this would indicate the presence of a borrowed size effect from Seoul. At the regional scale, we used the variable *DIST_CENTER*—the distance from each regional first city—to investigate whether borrowed size arises from

geographical proximity at the regional level. If we compare the statistical significance of the coefficients for *DIST_SEOUL* and *DIST_CENTER*, we can assess which scale of geographical proximity exerts a stronger influence on borrowed size within a national urban system characterised by strong primacy.

Descriptions of all variables are presented in Table 1. All data were collected as of 2022 except for GRDP (i.e. in this case, 2020) and at the spatial scale of city and county.¹

Regression models and estimation methods –

The regression models to be estimated are constructed to investigate whether a small- and medium-sized city's position in the urban hierarchy and/or its geography (distance to large cities) makes any difference in generating borrowed size effects. For this purpose, two sets of equations were established for the cases of two groups (second- and third-tier cities) and four-group classifications (upper and lower second-tier cities and upper and lower third-tier cities), each of which had three individual equations. Equations (1–3) describe the two-group case, whereas Equations (4–6) describe the four-group case. *i* represents 152 individual small- and medium-sized cities in the Korean urban system:

$$\text{Functions/performance}_i = \alpha + \beta_1 \text{SECOND}_i + \beta_3 \text{DIST_SEOUL}_i + \varepsilon_i \quad (1)$$

$$\text{Functions/performance}_i = \alpha + \beta_2 \text{REG_SECOND}_i + \beta_4 \text{DIST_CENTER}_i + \varepsilon_i \quad (2)$$

$$\text{Functions/performance}_i = \alpha + \beta_1 \text{SECOND}_i + \beta_2 \text{REG_SECOND}_i + \beta_3 \text{DIST_SEOUL}_i + \beta_4 \text{DIST_CENTER}_i + \varepsilon_i \quad (3)$$

$$\text{Functions/performance}_i = \alpha + \beta_1 \text{UP_SECOND}_i + \beta_2 \text{LOW_SECOND}_i + \beta_3 \text{UP_THIRD}_i + \beta_7 \text{DIST_SEOUL}_i + \varepsilon_i \quad (4)$$

$$\begin{aligned} \text{Functions/performance}_i = & \alpha + \beta_4 \text{REG_UP_SECOND}_i \\ & + \beta_5 \text{REG_LOW_SECOND}_i \\ & + \beta_6 \text{REG_UP_THIRD}_i \\ & + \beta_8 \text{DIST_CENTER}_i + \varepsilon_i \end{aligned} \quad (5)$$

$$\begin{aligned} \text{Functions/performance}_i = & \alpha + \beta_1 \text{UP_SECOND}_i \\ & + \beta_2 \text{LOW_SECOND}_i \\ & + \beta_3 \text{UP_THIRD}_i \\ & + \beta_4 \text{REG_UP_SECOND}_i \\ & + \beta_5 \text{REG_LOW_SECOND}_i \\ & + \beta_6 \text{REG_UP_THIRD}_i \\ & + \beta_7 \text{DIST_SEOUL}_i \\ & + \beta_8 \text{DIST_CENTER}_i + \varepsilon_i \end{aligned} \quad (6)$$

Equations (1) and (4) focus on the relationship between the hierarchical and geographical variables and borrowed size effects at the national scale. Therefore, positions were classified in the national urban system, and the distance to large cities was replaced by the distance to Seoul, Korea's largest city. Equations (2) and (5) represent analyses at the regional scale. Here, the positions are classified based on the individual regional urban systems in each province. Similarly, the distance to large cities is measured as the distance to the largest cities in each province, as mentioned earlier. Finally, Equations (3) and (6) incorporate the national and regional explanatory factors into a single equation. The estimated coefficients of these two equations enable us to discern which of the factors between position in the urban hierarchy and distance is more influential in generating borrowed size effects in small- and medium-sized cities in Korea. Furthermore, they help us determine which spatial scale of urban systems (national or regional) is more significant in driving positive spillover effects from large cities. The regression equations were estimated using OLS with cluster-robust standard errors by province to avoid heteroscedasticity and autocorrelation.

THE PRIMATE CITY SEOUL AND THE CHARACTERISTICS OF THE KOREAN URBAN SYSTEM

Seoul is not only the capital of Korea but also the dominant centre of the Korean economy. The city is characterised as a primate city in the Korean urban system with a primate distribution. Busan, the second-largest city in Korea, has less than half the population of Seoul. With the presence of Seoul, the Korean urban system is characterised as an urban system with a very strong urban hierarchy. Figure 3 shows the rank-size distribution of the Korean urban system for 2022. The blue line represents the conventional rank-size distribution with a very steep slope near the top-ranked city (Seoul), which tapers down as the rank decreases, suggesting a typical urban system with a very strong urban hierarchy. A more interesting observation was found in the rank-size distribution, where the rank was derived from the distance from Seoul (red line). Although a few large cities in the country are scattered in distance, including the second-largest city, Busan, which is ranked 154th in terms of distance from Seoul, the overall distribution trend shows a downward slope as the rank decreases. This suggests that the urban hierarchy is structured not only in terms of a city's population size but also in terms of the distance from the city to Seoul (Sohn 2012). In other words, it is important to consider the geographical aspects of a city as well as its size to understand the Korean urban hierarchy.

The fate of small- and medium-sized cities in the Korean urban system is twofold. If a more balanced city size distribution in the urban system is helpful for inducing the borrowed size effect, as can be seen in many European cities, the strong urban hierarchy of the Korean urban system, where there is a gap between large cities, including Seoul, and other small- and medium-sized cities, may result in the agglomeration shadow effect on the latter cities. However, if the positive spillover effect from the large city who acts as an anchor of the urban system is more prevalent, especially among adjacent cities,

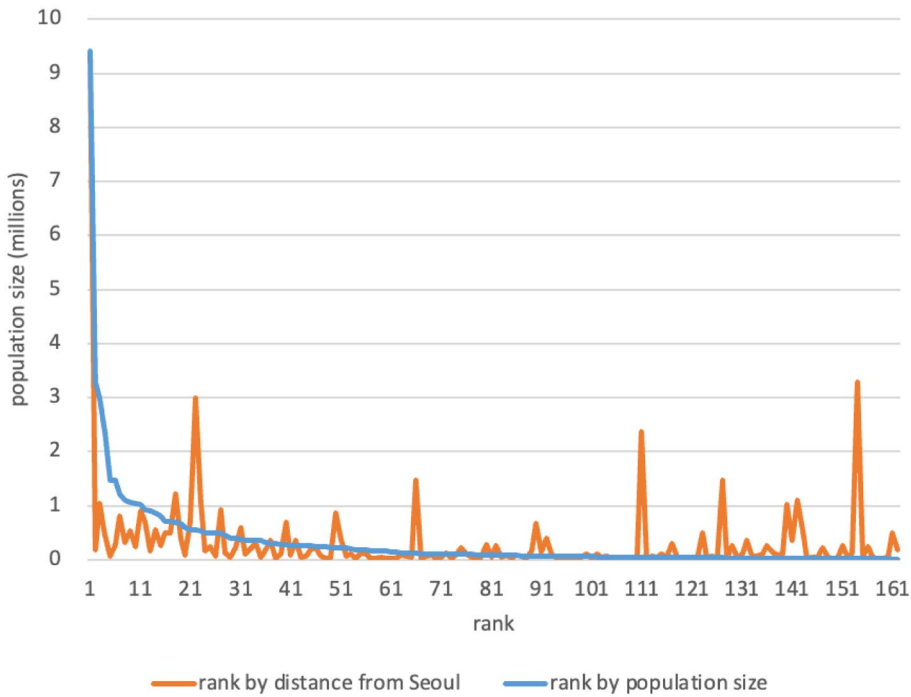


Figure 3. Rank-size distributions of national urban systems in Korea.

as is the case of London (Phelps *et al.* 2001), the Korean urban system is likely to show a borrowed size effect on small- and medium-sized cities, especially those adjacent to the top-ranked city, Seoul. To examine the direction of the relationship between large cities and small- and medium-sized cities, the next section reports the results of the regression analysis on borrowed size effects.

ESTIMATED IMPACTS OF URBAN HIERARCHY AND GEOGRAPHY ON BORROWED SIZE EFFECTS

Tables 2–4 present the regression results on how cities' positions in the urban hierarchy and geographic space influence the extent of borrowed size effects. In these analyses, cities are classified into two groups based on urban hierarchy: second- and third-tier cities. All coefficients are standardised for comparison across models.

Model 1 examines the national context, including two explanatory variables: a dummy

for second-tier cities (SECOND) and the distance to Seoul (DIST_SEOUL). The second-tier city dummy is positively significant in most cases, suggesting that cities with higher hierarchical positions are more likely to benefit from borrowed size effects. However, per capita GRDP (INCOME) is an exception; this index does not differ significantly between second- and third-tier cities. The distance to Seoul (DIST_SEOUL) exhibits a clear distance decay effect in five indices: advanced producer services (BUSINESS), creative industries (CULTURE), IT, patents (INNOVATE), and startups (VITALITY), with borrowed size effects decreasing as distance increases. Nevertheless, the second-tier city dummy usually has stronger explanatory power than distance to Seoul, highlighting that hierarchical position is a more crucial factor in borrowed size effects at the national scale.

Model 2 shifts the focus to the regional scale, incorporating second-tier cities in the regional hierarchy (REG_SECOND) and distance to the regional economic centre (DIST_CENTER). As in the national model, the regional second-tier

Table 2. *Estimation results for two-level urban hierarchy (Model 1).*

Variables	SECOND	DIST_SEOUL	Adj- R^2
BUSINESS	0.350 (0.07)***	-0.233 (0.11)*	0.221
CONSUME	0.296 (0.08)***	-0.139 (0.10)	0.123
CULTURE	0.334 (0.04)***	-0.274 (0.11)*	0.237
TRANSPORT	0.303 (0.04)***	0.007 (0.04)	0.078
IT	0.264 (0.09)**	-0.347 (0.14)*	0.241
INCOME	-0.029 (0.11)	-0.107 (0.12)	-0.003
INNOVATE	0.258 (0.06)***	-0.215 (0.08)**	0.138
VITALITY	0.319 (0.09)***	-0.305 (0.11)**	0.251

Cluster-robust standard error in parentheses (cluster: province).

* $p < 0.05$;

** $p < 0.01$;

*** $p < 0.001$.

Table 3. *Estimation results for two-level urban hierarchy (Model 2).*

Variables	REG_SECOND	DIST_CENTER	Adj- R^2
BUSINESS	0.540 (0.20)**	-0.148 (0.12)	0.333
CONSUME	0.560 (0.19)**	-0.121 (0.10)	0.343
CULTURE	0.460 (0.17)**	-0.185 (0.15)	0.269
TRANSPORT	0.245 (0.07)**	-0.053 (0.04)	0.055
IT	0.409 (0.22)	-0.213 (0.17)	0.233
INCOME	0.082 (0.09)	0.079 (0.10)	-0.003
INNOVATE	0.444 (0.21)*	-0.128 (0.08)	0.223
VITALITY	0.477 (0.19)*	-0.220 (0.13)	0.303

Cluster-robust standard error in parentheses (cluster: province).

* $p < 0.05$;

** $p < 0.01$;

dummy is positively associated with most borrowed size indices except for INCOME and information technology (IT). The insignificance of the IT index at the regional level suggests that

some knowledge-intensive activities may require a higher population threshold than that of regional second-tier cities. In contrast to Model 1, the distance to the regional centre shows little

Table 4. Estimation results for two-level urban hierarchy (Model 3).

Variables	SECOND	REG_SECOND	DIST_SEOUL	DIST_CENTER	Adj- <i>R</i> ²
BUSINESS	0.122 (0.07)	0.478 (0.22)*	−0.246 (0.12)*	−0.058 (0.05)	0.408
CONSUME	0.040 (0.09)	0.536 (0.22)*	−0.152 (0.11)	−0.071 (0.05)	0.360
CULTURE	0.145 (0.09)	0.390 (0.20)	−0.275 (0.12)*	−0.082 (0.07)	0.368
TRANSPORT	0.232 (0.06)***	0.149 (0.07)*	0.003 (0.05)	−0.020 (0.02)	0.084
IT	0.086 (0.08)	0.358 (0.23)	−0.342 (0.15)*	−0.101 (0.09)	0.353
INCOME	−0.061 (0.15)	0.100 (0.12)	−0.141 (0.11)	0.112 (0.12)	0.001
INNOVATE	0.064 (0.07)	0.407 (0.23)	−0.225 (0.10)*	−0.053 (0.05)	0.270
VITALITY	0.112 (0.07)	0.417 (0.2)*	−0.300 (0.12)*	−0.116 (0.06)	0.405

Cluster-robust standard error in parentheses (cluster: province).

**p* < 0.05;

****p* < 0.001.

explanatory power. Overall, regional borrowed size effects appear to be shaped by hierarchical position rather than geographic proximity. For example, this implies that cities distant from the largest city in the region but with a high urban hierarchy have stronger influences on borrowed size effects than those close to the largest city but with a low hierarchy.

Model 3 incorporates all four positional variables (SECOND, REG_SECOND, DIST_SEOUL and DIST_CENTER). Here, the significance of the national second-tier dummy weakens, whereas the regional second-tier dummy retains more explanatory power. The distance to Seoul continues to show significant distance decay effects across the five indices, even after considering the influences of other positional variables, reaffirming the strong centripetal force of Seoul in shaping urban dynamics. The distance to regional centres remains largely insignificant.

Comparing the four variables, distance to Seoul exhibits the largest number of significant coefficients (five indices), followed by the regional second-tier dummy (four indices). This indicates that, when both hierarchy and geography are considered, proximity to Seoul exerts the most consistent influence on the borrowed

size effects of small- and medium-sized cities in Korea. Thus, Seoul strongly influences urban economic activities in Korean cities.

Tables 5–7 further refine the analysis by disaggregating the urban hierarchy into four categories: upper and lower second-tier cities and upper and lower third-tier cities.

The first model (Model 4) in Table 5 revisits the national scale using this finer classification. The lower third-tier city dummy (LOW_THIRD) is the reference and the coefficients of the other three dummy variables are estimated. The results show that only the upper second-tier cities (UP_SECOND) consistently benefit from borrowed size effects across most indices, except for INCOME. Lower second-tier (LOW_SECOND) and upper third-tier (UP_THIRD) cities show limited and mixed effects, indicating that substantial borrowed size benefits are concentrated in the top segment of small- and medium-sized cities. Distance to Seoul maintains its distance decay effect in the five indices, as seen in the earlier models. Comparing the variables, the upper second-tier city dummy and distance to Seoul are the most influential, suggesting that either a high hierarchical status or proximity to Seoul is the key to benefiting from borrowed size effects.

Table 5. Estimation results for four-level urban hierarchy (Model 4).

Variables	UP_SECOND	LOW_SECOND	UP_THIRD	DIST_SEOUL	Adj- R^2
BUSINESS	0.608 (0.10)***	0.077 (0.03)**	0.017 (0.02)	-0.166 (0.05)**	0.411
CONSUME	0.558 (0.08)***	0.013 (0.02)	-0.004 (0.01)	-0.072 (0.06)	0.320
CULTURE	0.584 (0.12)***	0.056 (0.05)	0.002 (0.02)	-0.207 (0.06)***	0.430
TRANSPORT	0.446 (0.08)***	0.117 (0.01)**	0.002 (0.01)	0.048 (0.06)	0.144
IT	0.523 (0.15)***	-0.010 (0.05)	-0.006 (0.03)	-0.281 (0.08)***	0.430
INCOME	-0.008 (0.13)	0.035 (0.12)	0.080 (0.08)	-0.109 (0.11)	-0.011
INNOVATE	0.478 (0.09)***	0.032 (0.03)	0.011 (0.02)	-0.159 (0.03)***	0.267
VITALITY	0.624 (0.13)***	0.028 (0.03)	0.028 (0.02)	-0.230 (0.05)***	0.491

Cluster-robust standard error in parentheses (cluster: province).

** $p < 0.01$;

*** $p < 0.001$.

Table 6. Estimation results for four-level urban hierarchy (Model 5).

Variables	REG_UP_ SECOND	REG_LOW_SECOND	REG_UP_ THIRD	DIST_ CENTER	Adj- R^2
BUSINESS	0.633 (0.26)*	0.135 (0.05)**	0.117 (0.04)**	-0.145 (0.12)	0.410
CONSUME	0.580 (0.21)**	0.150 (0.10)	0.040 (0.03)	-0.128 (0.09)	0.350
CULTURE	0.548 (0.21)**	0.155 (0.06)*	0.110 (0.06)	-0.177 (0.15)	0.335
TRANSPORT	0.289 (0.07)***	0.221 (0.18)	0.012 (0.04)	-0.027 (0.06)	0.095
IT	0.493 (0.28)	0.083 (0.05)	0.091 (0.07)	-0.213 (0.17)	0.285
INCOME	0.110 (0.06)	0.035 (0.11)	0.091 (0.10)	0.085 (0.11)	-0.008
INNOVATE	0.554 (0.28)	0.096 (0.03)**	0.071 (0.03)*	-0.125 (0.08)	0.307
VITALITY	0.52 (0.24)*	0.169 (0.08)*	0.128 (0.08)	-0.214 (0.13)	0.326

Cluster-robust standard error in parentheses (cluster: province).

* $p < 0.05$;

** $p < 0.01$;

*** $p < 0.001$.

Model 5 applies the same hierarchical specifications at a regional scale. Here, upper regional second-tier cities (REG_UP_SECOND) show positive influences on the

five indices, whereas lower second-tier (REG_LOW_SECOND) and upper third-tier (REG_UP_THIRD) cities in the region display slightly limited effects. Notably, advanced producer

Table 7. Estimation results for four-level urban hierarchy (Model 6).

Variables	UP_ SECOND	LOW_ SECOND	UP_ THIRD	REG_ UP_ SECOND	REG_ LOW_ SECOND	REG_ UP_ THIRD	DIST_ SEOUL	DIST_ CENTER	Adj- <i>R</i> ²
BUSINESS	0.260 (0.15)	-0.029 (0.11)	-0.022 (0.05)	0.467 (0.30)	0.092 (0.08)	0.072 (0.06)	-0.236 (0.14)	-0.023 (0.04)	0.522
CONSUME	0.247 (0.19)	-0.068 (0.09)	-0.024 (0.05)	0.421 (0.25)	0.107 (0.13)	0.004 (0.05)	-0.133 (0.14)	-0.038 (0.04)	0.416
CULTURE	0.309 (0.14)*	-0.031 (0.13)	-0.031 (0.06)	0.350 (0.23)	0.101 (0.09)	0.048 (0.08)	-0.262 (0.12)*	-0.033 (0.05)	0.487
TRANSPORT	0.427 (0.14)**	0.129 (0.07)	0.036 (0.05)	0.048 (0.12)	0.088 (0.18)	-0.102 (0.10)	0.025 (0.09)	0.038 (0.05)	0.144
IT	0.285 (0.14)*	-0.082 (0.13)	-0.040 (0.07)	0.304 (0.30)	0.047 (0.08)	0.053 (0.08)	-0.315 (0.14)*	-0.059 (0.06)	0.470
INCOME	-0.196 (0.12)	-0.066 (0.14)	0.041 (0.05)	0.229 (0.07)***	0.122 (0.09)	0.142 (0.11)	-0.190 (0.10)*	0.114 (0.11)	0.001
INNOVATE	0.140 (0.15)	-0.066 (0.11)	-0.024 (0.05)	0.458 (0.32)	0.090 (0.07)	0.057 (0.06)	-0.226 (0.12)	-0.026 (0.04)	0.371
VITALITY	0.372 (0.12)**	-0.065 (0.11)	-0.015 (0.06)	0.286 (0.26)	0.107 (0.08)	0.067 (0.08)	-0.274 (0.12)*	-0.057 (0.04)	0.524

Cluster-robust standard error in parentheses (cluster: province).

**p* < 0.05;

***p* < 0.01;

****p* < 0.001.

services (BUSINESS) show positive associations across all three higher hierarchy categories, suggesting that this sector is less sensitive to hierarchical status than others. Unlike the results in Model 4 with four urban hierarchy categories in the national context, cities in the lower urban hierarchy still have some meaningful influences on borrowed size indices. However, such influences decrease as the urban hierarchical level decreases. In other words, at the national level, a substantial increase in the influence of borrowed size effects is possible only in cities with relatively higher positions in the hierarchy. At the regional level, however, small- and medium-sized cities can enjoy borrowed size effects even if they slightly move up in the urban hierarchy. The distance to regional centres remains mostly insignificant, reinforcing that hierarchy matters more than geography at the regional scale.

Model 6 combines all the national and regional hierarchies and geographical variables. In this comprehensive model, only national upper second-tier cities (UP_SECOND) consistently show positive and significant impacts on borrowed size effects, particularly for

CULTURE, TRANSPORT, IT and VITALITY. Regional hierarchical variables lose significance when national variables are included, suggesting that the national urban hierarchy outweighs regional dynamics. Regarding geography, only the distance to Seoul has a significant and positive influence across multiple indices, while the distance to regional centres remains negligible.

In summary, when both hierarchical and geographic positions are considered, borrowed size effects in Korea's small- and medium-sized cities are predominantly shaped by two factors: being in the top tier of the national urban hierarchy and proximity to Seoul. When comparing the hierarchy and geography variables, national upper second-tier cities and distance to Seoul seem to have similar levels of influence. Interestingly, the significance of geographic distance diminishes when the hierarchical classification is refined. This implies that, at a coarse level, geography appears to be influential because urban hierarchical characteristics are not well represented; however, once the hierarchical structure is more precisely

specified, the importance of hierarchy becomes more apparent.

DISCUSSION

Figure 4 summarises the empirical findings of the previous section. It compares the relative influences of urban hierarchy and geographic position on borrowed size effects in small- and medium-sized cities. Based on these findings, three key points of discussion were identified.

The first concerns a city’s position in the urban hierarchy. Under the two-level

classification (second- and third-tier cities), a higher hierarchical status is positively associated with the level of urban functions and performance, both nationally and regionally. In other words, cities in higher positions tend to benefit more from these effects. However, when the hierarchy is disaggregated into four levels (upper and lower second-tier and upper and lower third-tier cities), the pattern changes. At the national level, only the top segment (upper second-tier cities) shows meaningful positive effects. By contrast, at the regional level, although the top segment has the strongest influence, even lower-tier cities display some positive effects, albeit

Position in hierarchy effects

	National urban system	Regional urban system
Two-level	National second-tier cities (1)	Regional second-tier cities (4)
Four-level	National upper-second-tier cities (4)	Regional upper-second-tier cities (1)

Position in geography effects

	Distance to Seoul	Distance to regional economic center
Two-level	Distance to Seoul (5)	Distance to regional economic center (0)
Four-level	Distance to Seoul (4)	Distance to regional economic center (0)

Comparison between hierarchy and geography

	Position in hierarchy	Position in geography
Two-level	Regional second-tier cities (4)	Distance to Seoul (5)
Four-level	National upper-second-tier cities (4)	Distance to Seoul (4)

Figure 4. Generalisation of the influences of factors on borrowed size indicators. (1) The information here shows which of the two variables in comparison contributes more to the dependent variable in the eight regression models. The number in parentheses indicates the number of times the corresponding variable is observed as significant among eight regressions. The variable with a higher number is shown in bold. (2) The results of the two- and four-level hierarchy models are derived from Models 3 and 6, respectively (Tables 4 and 7).

weaker. This suggests that, in terms of positive influences on borrowed size effects, the national urban system has a stronger hierarchy than the regional ones, meaning that the difference between city groups is smaller in the regional urban hierarchy.

When both national and regional hierarchy variables are included in one model, the results differ according to the classification scheme. In the two-level model, regional hierarchical status appears more influential than national status. This implies that a city's position in its regional hierarchy may matter more for borrowed size effects than its national rank. This could explain why some medium-sized cities in the Seoul Metropolitan Area, although ranked higher nationally, may not benefit if they occupy lower positions regionally.

In contrast, the four-level model shows that only national upper second-tier cities have a significant impact, whereas regional hierarchical variables lose their influence. This suggests that only cities at the top of the hierarchy can benefit substantially from borrowed size effects. For lower-tier cities, simply climbing the hierarchy may not guarantee positive effects—an outcome that may be discouraging from a policy perspective.

Taken together, these contrasting results point to a dual dynamic in Korea's urban system. This pattern reflects the coexistence of two institutional tendencies: while the central government continues to prioritise investments in higher-order cities to maximise returns, it simultaneously pursues balanced development projects and allocates resources to promote regional equity. This is consistent with studies on China that highlight the importance of urban hierarchy and borrowed administrative power in the opening of high-speed rail stations and the industrial upgrading process (Wang *et al.* 2022b; Shi *et al.* 2025). Although our model cannot disentangle the effects of these institutional factors related to hierarchical proximity, our findings can be interpreted as capturing the combined influence of both tendencies.

The second aspect concerns the role of geographic distance. In all models, the distance from Seoul shows a strong negative association with borrowed size effects, revealing a clear distance decay pattern. In contrast, the distance from regional core cities is largely

insignificant, indicating that spillover effects originate mainly from Seoul rather than from regional centres. This spatial asymmetry reflects Seoul's overwhelming dominance in Korea's urban system—not only in size but also in economic, political and cultural influence. Korea's compact territory and transport network oriented toward the capital further amplify these effects, making proximity to Seoul a decisive factor for realising borrowed size benefits. Similar spatial patterns have been observed in other primate city systems, such as in the UK, where distance from London weakens spillover effects (Phelps *et al.* 2001), and in Japan, where greater accessibility to Tokyo via high-speed rail has promoted diffusion from the capital (Li & Xu 2018).

Finally, a comparison of the two factors reveals different conclusions depending on the classification scheme. In the two-level model, the distance to Seoul exerts a stronger influence than the regional hierarchy. This implies that spatial proximity to the primate city is a more efficient driver of borrowed size effects than hierarchical position. However, when the hierarchy is specified in four levels, the influence of the national hierarchy becomes comparable to that of distance. This suggests that a coarse classification may exaggerate the role of geography by failing to distinguish between genuine hierarchical differences. By contrast, a more refined classification reveals that both hierarchy and distance matter equally.

Overall, the findings suggest that small- and medium-sized cities in Korea can benefit from borrowed size effects under two conditions: either by holding a top position in the urban system of small- and medium-sized cities or by being geographically close to Seoul.

These findings offer several policy implications that directly connect to its theoretical underpinnings, particularly in East Asian cities where government intervention in urban and regional development planning is relatively strong. According to the concepts of borrowed size, the positions of smaller cities within both geographical and hierarchical networks can either constrain or enhance their development trajectories.

First, although the physical distance to Seoul cannot be changed, the relative distance can be reduced by improving transport connectivity. In

the theoretical framework of urban networks, such improvements can expand the functional catchment areas of small- and medium-sized cities, enabling them to borrow agglomeration benefits from Seoul. Even though it is impossible to change the absolute location (or site) of an entire city, it is possible to change its relative location (or situation). Such efforts should be accompanied by policies that foster functional complementarity to avoid the centralisation of functions in Seoul. Without functional complementarity, improved accessibility may merely increase the agglomeration shadows. Although Korea has implemented a local autonomy system for about 30 years, local governments have continued to rely heavily on the national government projects and budgets and have not yet sufficiently developed functional complementarities with larger cities. Recently, the Presidential Committee for Balanced National Development was reorganised and renamed the 'Presidential Committee for Decentralization and Balanced Development,' reflecting efforts to expand the responsibilities and authority of local governments. In this context, local governments need to further specialise their regional industries and strengthen functional complementarities with Seoul and other major cities.

Second, efforts to simply increase the population of small and medium-sized cities need to be reconsidered. Due to the concentration of population in the Seoul Metropolitan Area, these cities are experiencing rapid urban shrinkage and population decline. Without proactive policy interventions, even the provision of basic services could become unsustainable. However, given Korea's low fertility rate, attempts to expand the permanent resident population are likely to result in zero-sum outcomes. As our findings show, geographical and hierarchical proximity to Seoul is important for urban functions and performance. For small and medium-sized cities located far from Seoul, this suggests that instead of relying on borrowed size from the capital, policies to attract foreign immigrants and to leverage mobile populations such as commuters, part-time residents, business travellers and workation participants could be crucial for sustaining urban vitality.

Finally, the findings underscore the need to consider intercity interactions in urban

policy. Rather than merely identifying functional deficiencies in smaller cities, it is essential to examine how they can leverage the advantages of larger neighbouring cities through spatial and hierarchical networks. This point is particularly crucial in the Chinese context, where administrative hierarchies largely determine cities' capacity for resource allocation (Shi *et al.* 2025). Under Korea's megaregion policy, local governments are encouraged to establish collaborative zones linking major cities with surrounding smaller ones and to formulate joint regional master plans. Advancing this policy effectively requires a deeper understanding of intercity interactions and the roles of geographical and hierarchical networks.

CONCLUSION

This study examined whether and how small- and medium-sized cities can generate borrowed size effects within Korea's highly hierarchical urban system. Specifically, it investigated the roles of two key factors: a small- and medium-sized city's position in the urban hierarchy and geography. Hierarchical positions were measured using categorical variables based on national and regional rankings, whereas geographic positions were captured by the distance to the primate city and regional core cities. The level of urban functions and performance was measured using eight indices to assess the existence of borrowed size effects across 152 small- and medium-sized cities.

The empirical analysis yielded three key findings. First, the hierarchical position matters, but its influence varies depending on the classification scheme. Under a simple two-tier classification, both national and regional second-tier cities benefit from borrowed size effects, with regional hierarchy being more influential. However, when the hierarchy is disaggregated into four levels, only the upper second-tier cities at the national level show meaningful positive effects. Second, the distance to Seoul consistently shows a significant negative association with borrowed size effects across all models, indicating a clear distance decay effect, whereas the distance

to regional core cities has no meaningful influence. Third, when comparing the hierarchy and distance variables, the proximity to Seoul proves to be the most influential factor, especially when a simple hierarchical classification is used. However, with finer classification, the influence of the national urban hierarchy becomes comparable to that of geographic proximity.

These findings suggest that in Korea, small and medium-sized cities can benefit from borrowed size effects only when they are geographically close to Seoul or occupy higher positions in the national urban hierarchy. Yet these conditions are difficult to meet because of Seoul's entrenched dominance and the rigid hierarchical structure of the urban system. The geographical distance to Seoul cannot be altered, and few smaller cities can attract enough population to move up the hierarchy unless they already possess major economic or administrative functions. Nonetheless, improving transportation connectivity to reduce relative distance offers a practical policy option for enabling smaller cities to borrow agglomeration benefits despite these structural constraints. However, accessibility improvements alone may intensify the concentration of functions in the primary city and expand agglomeration shadows. Thus, regional policy should aim to foster functional complementarity and strengthen intercity interactions through spatial and hierarchical networks.

This study contributes to the broader discussion of agglomeration and network effects by shedding light on the structural conditions under which borrowed size effects emerge. However, we were unable to distinguish the individual processes of hierarchical and geographical proximity as separate explanatory variables and thus could not determine which exerted the strongest influence on borrowed size. This limitation underscores the need for further investigation. Future research could extend this work by examining temporal changes in these effects, the capacity of individual cities to capitalise on them, and how different types of network connections—such as commuting, logistics and accessibility—shape the geography of borrowed size effects in Korea and beyond.

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CONFLICT OF INTEREST

The authors declare that they have no competing interests.

Endnote

- ¹ The administrative divisions of Korea are divided into three levels: (1) metropolitan cities and provinces; (2) cities, counties, and districts and (3) neighbourhoods and townships. In this study, data were collected at the second level, but the districts were not included in the analysis. This is because the districts are only parts (or lower units) of the metropolitan cities rather than a city of its own; thus, the use of the district level may not be appropriate for the purposes of this study to examine the relationships between administratively independent cities.

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